JavaWorkeSpaces - knscs03/src/Abstract_Class_Assignment/ElectricCar.java - Eclipse IDE

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Vehicle.java

```java
package Abstract_Class_Assignment;
public abstract class Vehicle {
    String brand;
    double speed;
    double fuelCapacity;
    public Vehicle(String brand, double speed,
            double fuelCapacity) {
        super();
        this.brand = brand;
        this.speed = speed;
        this.fuelCapacity = fuelCapacity;}
    public void displayInfo() {
        System.out.printf("Brand: %-15s | Speed: %.0f km/h%n",
                brand, speed);}
    public abstract double calculateFuelEfficiency();
    public abstract String makeSound();
    public abstract String getVehicleType();
}
```

Car.java

```java
package Abstract_Class_Assignment;
public class Car extends Vehicle {
    public Car(String brand, double speed,
            double fuelCapacity) {
        super(brand, speed, fuelCapacity);}
    @Override
    public double calculateFuelEfficiency() {
        return (fuelCapacity / speed) * 10;
    }
    @Override
    public String makeSound() {
        return "Vroom Vroom";
    }
    @Override
    public String getVehicleType() {
        return "Car";
    }
}
```

Truck.java

```java
package Abstract_Class_Assignment;
public class Truck extends Vehicle{
    public Truck(String brand, double speed,
            double fuelCapacity) {
        super(brand, speed, fuelCapacity);}
    @Override
    public double calculateFuelEfficiency() {
        return (fuelCapacity / speed) * 5;
    }
    @Override
    public String makeSound() {
        return "Horn Blows!";
    }
    @Override
    public String getVehicleType() {
        return "Truck";
    }
}
```

Bike.java

```java
package Abstract_Class_Assignment;
public class Bike extends Vehicle{
    public Bike(String brand, double speed, double fuelCapacity)
    {super(brand, speed, fuelCapacity);}
    @Override
    public double calculateFuelEfficiency() {
        return (fuelCapacity / speed) * 15;
    }
    @Override
    public String makeSound() {
        return "Vroom Vroom";
    }
    @Override
    public String getVehicleType() {
        return "Bike";
    }


}
```

ElectricCar.java

```java
package Abstract_Class_Assignment;
public class ElectricCar extends Vehicle{
    public static final double KWH_PER_LITRE_EQUIVALENT = 8.9;
    public ElectricCar(String brand, double speed,
            double batteryCapacityKWh)
    {super(brand, speed, batteryCapacityKWh);}
    @Override
    public double calculateFuelEfficiency() {
        return (fuelCapacity * KWH_PER_LITRE_EQUIVALENT) / speed * 10;}
    @Override
    public String makeSound() {
        return "Silent Hum";}
    @Override
    public String getVehicleType() {
        return "Electric Car";}
}
```

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JavaWorkeSpaces - knscs03/src/Abstract_Class_Assignment/Main.java - Eclipse IDE

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Main.java

```java
package Abstract_Class_Assignment;
import java.util.ArrayList;
public class Main {
    private static void printVehicleDetails(Vehicle v) {
        v.displayInfo();
        System.out.printf("Vehicle Type      : %s%n",   v.getVehicleType());
        System.out.printf("Fuel Efficiency   : %.2f km/l%n", v.calculateFuelEfficiency());
        System.out.printf("Sound             : %s%n",   v.makeSound());
        System.out.println("-".repeat(45));
    }
    public static void main(String[] args) {
        Car   car   = new Car  ("Toyota",        80, 50);
        Bike  bike  = new Bike ("Royal Enfield", 60, 30);
        Truck truck = new Truck("Tata Truck",     40, 30);
        System.out.println("=".repeat(45));
        System.out.println("   VEHICLE MANAGEMENT SYSTEM");
        System.out.println("=".repeat(45));
        car.displayInfo();
        System.out.printf("Fuel Efficiency : %.2f km/l%n", car.calculateFuelEfficiency());
        System.out.printf("Sound           : %s%n%n",       car.makeSound());
        bike.displayInfo();
        System.out.printf("Fuel Efficiency : %.2f km/l%n", bike.calculateFuelEfficiency());
        System.out.printf("Sound           : %s%n%n",       bike.makeSound());
        truck.displayInfo();
        System.out.printf("Fuel Efficiency : %.2f km/l%n", truck.calculateFuelEfficiency());
        System.out.printf("Sound           : %s%n",         truck.makeSound());
        System.out.println("\n" + "=".repeat(45));
        System.out.println("   ALL VEHICLES (ArrayList iteration)");
        System.out.println("=".repeat(45));
        ElectricCar electricCar = new ElectricCar("Tesla Model 3", 120, 75);
        ArrayList<Vehicle> fleet = new ArrayList<>();
        fleet.add(car);
        fleet.add(bike);
        fleet.add(truck);
        fleet.add(electricCar);
        for (Vehicle v : fleet) {
            printVehicleDetails(v);
        }
    }

```

Console

<terminated> Main (2) [Java Application] C:\Users\Ganesh\softwares\eclipse-jee-2026-03-R-win32-x86_64\eclipse\plugins\org.eclipse

```
=============================================
   VEHICLE MANAGEMENT SYSTEM
=============================================
Brand: Toyota          | Speed: 80 km/h
Fuel Efficiency : 6.25 km/l
Sound           : Vroom Vroom

Brand: Royal Enfield   | Speed: 60 km/h
Fuel Efficiency : 7.50 km/l
Sound           : Vroom Vroom

Brand: Tata Truck      | Speed: 40 km/h
Fuel Efficiency : 3.75 km/l
Sound           : Horn Blows!

=============================================
   ALL VEHICLES (ArrayList iteration)
=============================================
Brand: Toyota          | Speed: 80 km/h
Vehicle Type      : Car
Fuel Efficiency   : 6.25 km/l
Sound             : Vroom Vroom
---------------------------------------------
Brand: Royal Enfield   | Speed: 60 km/h
Vehicle Type      : Bike
Fuel Efficiency   : 7.50 km/l
Sound             : Vroom Vroom
---------------------------------------------
Brand: Tata Truck      | Speed: 40 km/h
Vehicle Type      : Truck
Fuel Efficiency   : 3.75 km/l
Sound             : Horn Blows!
---------------------------------------------
Brand: Tesla Model 3   | Speed: 120 km/h
Vehicle Type      : Electric Car
Fuel Efficiency   : 55.63 km/l
Sound             : Silent Hum
---------------------------------------------
```

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